

Assignment 1: Linux Command-Line Basics

Course: Cybersecurity, Tutedude

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Topic: Orientation on a Linux target machine (pwd and whoami)

Overview

Before you can enumerate, exploit, or defend a Linux system, you need to answer two questions: **where am I?** and **who am I?** This assignment covers the two commands that answer them, `pwd` and `whoami`, and documents the output captured from the lab machine.

These are the first commands run after gaining a shell in almost every scenario, whether it is a CTF, a penetration test, or a SOC investigation on a suspicious host.

Objectives

- Identify the current working directory using a Linux command
- Identify the user account the session is running as
- Record the commands, outputs, and screenshot evidence

Lab Environment

Operating system	Ubuntu 24.04.3 LTS
Kernel	Linux 6.8.0-101-generic (x86_64)
Hostname	hacking
Logged-in user	student
Shell prompt	student@hacking:~\$
Session date	Wed, 23 Sep 2026

Task 1: Show the current directory

Question: What command shows the current directory?

Command

```
pwd
```

Output

```
/home/student
```

Explanation

`pwd` stands for **print working directory**. It prints the absolute path of the directory your shell is currently in, starting from the root (/).

The output `/home/student` is the home directory of the `student` user. This matches the `~` in the shell prompt (`student@hacking:~$`), since `~` is shorthand for the current user's home directory.

Useful options

Option	Behaviour
<code>pwd -L</code>	Logical path, keeps symbolic links as-is (default in Bash)
<code>pwd -P</code>	Physical path, resolves symbolic links to the real location

Why it matters in security: Knowing your exact location is essential when navigating an unfamiliar system, running scripts with relative paths, or locating files such as configs, logs, and credentials.

Task 2: Show the logged-in username

Question: What is the username you are logged in as?

Command

```
whoami
```

Output

```
student
```

Explanation

`whoami` prints the **effective username** of the user running the current shell. Here it returns `student`, a standard, unprivileged account. This tells us the session does not have administrator (root) rights.

Why it matters in security: Your privilege level determines what you can read, modify, and execute. Checking `whoami` is typically the first step in privilege-escalation work: a low-privileged user like `student` has to find a route to higher access, while a defender wants to confirm that services and users are running with the least privilege necessary.

Screenshot Evidence

```
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WELCOME TO THE CyberSecurity CHALLENGE
-----

TARGET IP ADDRESS: 2409:4091:8001:7e04:20c:29ff:fe06:2d66
WEB PORT: 8080

Login as 'student:student123' via SSH once you have credentials.
-----
hacking login: student
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 23 02:39:52 PM UTC 2026

System load:          0.06
Usage of /:            37.9% of 9.75GB
Memory usage:         17%
Swap usage:           0%
Processes:            259
Users logged in:      0
IPv4 address for ens33: 192.168.31.236
IPv6 address for ens33: 2409:4091:8001:7e04:20c:29ff:fe06:2d66

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

74 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

student@hacking:~$ pwd
/home/student
student@hacking:~$ whoami
student
student@hacking:~$
```

Figure 1: Login to the lab machine, followed by pwd returning /home/student and whoami returning student.

Observations from the Login Banner

The system banner shown at login also gives a quick look at the machine's state:

- **OS:** Ubuntu 24.04.3 LTS, running kernel 6.8.0-101-generic
- **Pending updates:** 74 updates were available and the package list was more than a week old. On a real system, unapplied patches increase the attack surface, and this is the kind of thing vulnerability management and SOC teams track.
- **Ubuntu Pro / ESM Apps:** Not enabled, so extended security maintenance for application packages is not active.
- **Network interface:** ens33 is the active interface, configured with both IPv4 and IPv6 addresses.

Key Takeaways

- 1 `pwd` shows **where** you are in the filesystem, and `whoami` shows **who** you are on the system.
- 2 Both are quick, safe, read-only commands that are the foundation of situational awareness on any Linux host.
- 3 Running as a non-root user (`student`) reflects the principle of least privilege.
- 4 Documenting commands with their output and screenshots makes work reproducible, which is a core habit in penetration testing and incident response.

Related Commands (Extra Practice)

These were not required for the assignment, but they build naturally on `pwd` and `whoami`:

Command	What it does
<code>id</code>	Shows your user ID, group ID, and all group memberships
<code>hostname</code>	Prints the name of the machine
<code>uname -a</code>	Displays kernel version and system architecture
<code>ls -la</code>	Lists all files, including hidden ones, with permissions
<code>cd ~</code>	Returns to your home directory from anywhere

Disclaimer: This work was completed in an authorized, purpose-built training environment provided by the course. It is shared for educational purposes only. Only run security tools and commands on systems you own or have explicit permission to test.